

Greenhouse Monitoring System



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Greenhouse Monitoring System Using Arduino

Objective

The primary objective of this project is to create an automated monitoring system for a greenhouse using Arduino. The system aims to monitor critical environmental parameters such as temperature, humidity, soil moisture, and light intensity. By maintaining optimal conditions, the system will enhance plant growth, reduce manual labor, and increase the overall efficiency of the greenhouse. The data collected will be used to automate environmental controls, ensuring that plants receive the ideal conditions for growth.

Theory

The greenhouse monitoring system is based on the principle of maintaining an optimal environment for plant growth. The system uses various sensors to measure key environmental parameters:

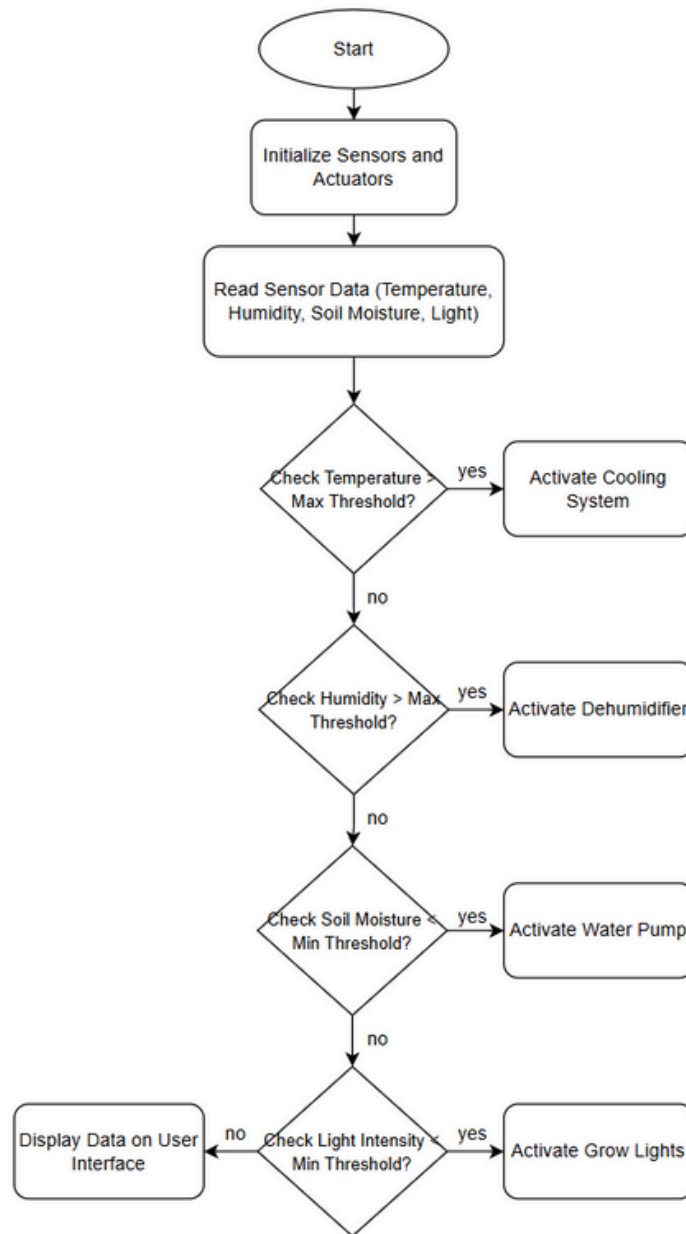
- **Temperature Sensor:** Measures the ambient temperature inside the greenhouse. The DHT11 sensor is commonly used for this purpose.
- **Humidity Sensor:** Monitors the relative humidity levels. The DHT11 sensor also provides humidity readings.
- **Soil Moisture Sensor:** Detects the moisture content in the soil. This sensor helps in determining when the plants need watering.
- **Light Sensor:** Measures the intensity of light available to the plants. An LDR (Light Dependent Resistor) is typically used for this purpose.

The Arduino microcontroller processes the data from these sensors and can trigger actuators such as fans, heaters, water pumps, and lights to maintain the desired conditions. The system operates based on predefined thresholds for each parameter.

Design

The design of the system includes the following components:

Flowchart :



Circuit Diagram :

The circuit diagram illustrates the connections between the Arduino, sensors, and actuators. Key components include:

Arduino Uno: The main microcontroller.

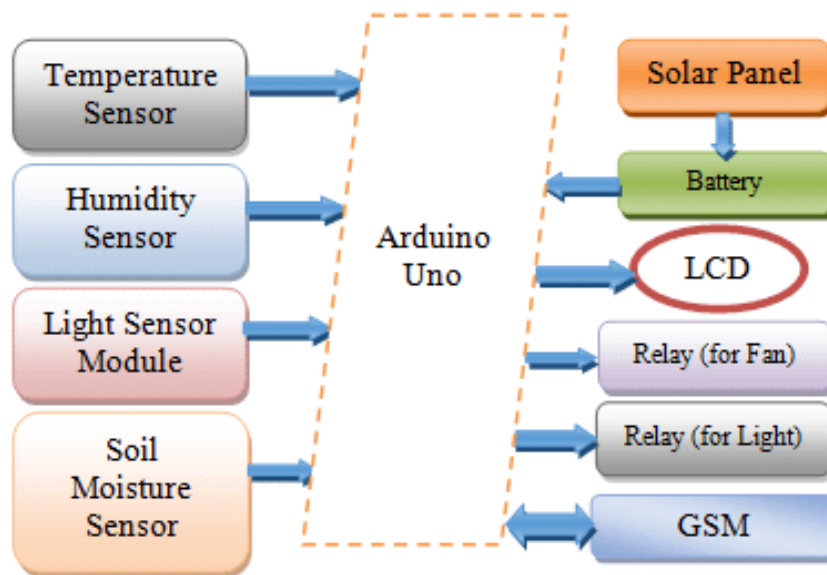
DHT11 Sensor: For temperature and humidity monitoring.

Soil Moisture Sensor: For soil moisture detection.

LDR: For light intensity measurement.

Relay Module: To control high-power devices like fans and water pumps.

LCD Display: To show real-time data.



Algorithm

The algorithm describes the logic used to process sensor data and control actuators. It includes:

Initialization of sensors and actuators.

Continuous reading of sensor data.

Comparison of sensor data with predefined thresholds.

Activation of actuators if any parameter is out of the desired range.

1. Read sensor values (A0-A2).
2. Convert raw ADC values to real units (°C, %).
3. Display on LCD.
4. Compare with thresholds:
5. Light < 20% → Light ON
6. Soil Moisture < 35% → Pump ON
7. Repeat in a loop.

Implementation

The implementation involves the following steps:

Hardware Setup

- Arduino Uno: Connect the sensors and actuators to the Arduino board as per the circuit diagram.
- DHT11 Sensor: Connect to analog pin A0 .
- Soil Moisture Sensor: Connect to analog pin A2.
- LDR: Connect to analog pin A1.
- Relay Module: Connect to digital pins 6, 7, 5 and 4.
- LCD Display: Connect to digital pins 13, 12, 11, 10, 9, and 8.

Software Development

- Arduino IDE: Write and upload the Arduino code to read sensor data and control actuators.
- Libraries: Use libraries such as DHT.h for the DHT11 sensor and LiquidCrystal.h for the LCD display.

Functional Modules

- Data Acquisition Module: Reads data from sensors.
- Control Module: Processes data and controls actuators.
- Display Module: Shows the current status on an LCD or sends data to a computer.

System Requirements

- Hardware: Arduino Uno, DHT11 (Temperature and Humidity Sensor), Soil Moisture Sensor, LDR (Light Sensor), Relay Module, LCD Display.
- Software: Arduino IDE, Libraries for sensors.

Debugging and Test-run

Testing: Each sensor and actuator is tested individually to ensure proper functioning.

Integration Testing: The entire system is tested to ensure all components work together seamlessly.

Debugging: Any issues in the code or hardware connections are identified and resolved.

Results Analysis

- Performance: The system effectively monitors and controls the greenhouse environment.
- Complexity: The algorithms used are efficient with low time and space complexity.
- Robustness: The system is reliable and can handle variations in environmental conditions

Parameter	Measured Value	Control Action
Temperature	25°C	None
Light Intensity	20%	Light ON
Soil Moisture	35%	Pump ON
Humidity	60%	None

Conclusion and Future Improvements

Conclusion

- The system successfully automates greenhouse monitoring.
- Low-cost & efficient compared to manual methods.
- Real-time adjustments improve plant growth conditions.

Future Improvements

- IoT Integration (Remote monitoring via Wi-Fi/Bluetooth).
- Solar Power for energy efficiency.
- Machine Learning for dynamic threshold adjustments.

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